

Atoms and Elements

Atom: Smallest particles that consists of protons, neutrons and electrons

Element: Same type of atoms combined with each other that cannot be chemically decomposed

Compound: Two or more different elements chemically combined

Molecule: Two or more elements physically combined.

* Elements in the same group have the same number of valency electrons, therefore share similar chemical properties.

* Elements in the same period have the same number of electron shell.

Metalloids: Substances that have the same characteristics of metals and some of non-metals.

Boron, Silicon, Germanium, Arsenic, Antimony, Tellurium, Astatine

→ Metals are to the left of the metalloids

→ Non-metals are to the right of the metalloids.

A mixture can be:

1- Homogeneous: One layer

2- Heterogeneous: Two or more layers

	Mass	Charge
Proton	1	1+
Neutron	1	0
Electron	negligible	1-

Atomic number: The number of protons in an atom, which equals the number of electrons in a neutral atom

Atomic mass: The number of protons and neutrons in the nucleus of an atom.

Elements in group 8 are called noble gases because they are inert (unreactive elements) as they have full outer shell: (eight valency electrons)

→ Helium is the only exception where it has only 2 electrons and is completely stable because the first shell can take only 2 electrons, while other noble gases are stable with 8 electrons.

→ Noble gases are colorless, odorless and tasteless, and they glow if electricity passes through them.

Isotopes: Elements that have the same number of protons but different number of neutrons.

→ Protium: Hydrogen with 1 proton, 1 electron, 0 neutrons

→ Deuterium: Hydrogen with 1 proton, 1 electron, 1 neutron

→ Tritium: Hydrogen with 1 proton, 1 electron, 2 neutrons

Why does C-14 have low abundance?

Because it is unstable (radioactive), so it eventually decomposes, releasing extra energy and radiation

Do isotopes have the same chemical properties?

Yes, because they have the same number of valency electrons.

Relative atomic mass: A_r : A value that is close to the number of atomic mass of the isotope with the highest percent of abundance.

$(\text{Atomic mass of first isotope} \times \text{its abundance}) + (\text{Atomic mass of second isotope} \times \text{its abundance})$

What is the reference atom in the periodic table?

$\frac{1}{12}$ of C-12

What are the uses of radioactive isotopes?

- 1- Carbon-dating: To detect the age of ancient remains by measuring C-14 radioactivity
- 2- Tracers: To detect leakage in oil or gas pipes
- 3- To treat cancer
- 4- To kill germs and bacteria

What are the dangers of radioactive isotopes?

- 1- Vomiting
- 2- Tiredness
- 3- Hair falling out
- 4- Gums bleeding
- 5- Causing cancer